Title: Dutton's Unified SEFI: A Single-Entity Field Interpretation of Electrons, Positrons, and Quantum Interference

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Abstract

Dutton's Unified SEFI proposes that all electrons and positrons are manifestations of a single underlying entity whose worldline spans all possible spacetime paths. The dense superposition of this worldline appears as a continuous field, reproducing the electron field of quantum field theory. Quantum interference arises from overlapping contributions of different segments of the entity's path structure, while measurement corresponds to selecting a subset of paths, producing a localized particle event. The model is formulated using relativistic worldline actions, Grassmann spin variables, electromagnetic gauge coupling, presence-density mapping, and the Dirac field in curved spacetime. SEFI preserves all predictions of quantum mechanics and quantum field theory while offering a unified ontological interpretation of particle identity and field behavior. Optional geometric and metaphysical extensions are clearly labeled and do not modify the core physics.

1. Introduction

Quantum mechanics and quantum field theory describe electrons as excitations of a field, yet the ontological nature of particle identity and self-interference remains conceptually unresolved. Dutton's Unified SEFI provides a unified interpretation in which all electrons and positrons are manifestations of a single underlying entity.

2. Core Postulates

P1 — Single Underlying Entity All electrons and positrons are manifestations of one fundamental entity whose worldline spans all possible spacetime paths.

P2 — Bidirectional Time Segments Forward-time segments appear as electrons; backward-time segments appear as positrons.

P3 — Dense Spacetime Weaving The entity's worldline is densely superposed across spacetime, producing an effective continuous field.

P4 — Emergent Field Appearance The observed electron field corresponds to the presence density generated by the entity's full path ensemble.

P5 — Interference as Path Overlap Quantum interference arises from overlapping contributions of different segments of the entity's worldline.

P6 — Measurement as Path Selection Measurement constrains the entity to a subset of its possible paths, producing a localized particle detection.

P7 — Ontological Unity of Particle Identity Identical particle properties arise because all electrons and positrons are local manifestations of the same underlying entity.

3. Mathematical Framework

3.1 Worldline Foundation

$$S_{\text{WL}} = -m \int d\,\tau \sqrt{-g_{\mu\nu}(\gamma)\dot{\gamma}^{\mu}\dot{\gamma}^{\nu}}$$

3.2 Spin Structure

$$S_{\text{spin}} = \int d\,\tau \left[\frac{i}{2}\Psi_{a}\dot{\Psi}^{a}\right]$$

3.3 Electromagnetic Gauge Coupling

$$S_{\text{EM}} = q \int d\,\tau\, \dot{\gamma}^{\mu} A_{\mu}\big(\gamma(\tau)\big)$$

3.4 Total Worldline Action

$$S_{\text{entity}} = S_{\text{WL}} + S_{\text{spin}} + S_{\text{EM}}$$

3.5 Presence Density

$$\rho(x) = \int \mathcal{D}\,[\gamma] e^{\frac{i}{\hbar}S[\gamma]} \delta^{(4)}\big(x - \gamma(\tau)\big)$$

3.6 Emergent Field

$$\psi(x) = \int d\,\tau\, \delta^{(4)}\big(x - \gamma(\tau)\big)$$

3.7 Dirac Field Structure

$$\mathcal{L}_{\text{Dirac}} = \bar{\psi}\big(i\gamma^{\mu} D_{\mu} - m\big)\psi$$

3.8 Time Reversal and Positrons

Backward-time segments correspond to positron manifestations.

3.9 Curved Spacetime Compatibility

$$D_{\mu} = \partial_{\mu} + \Omega_{\mu} + iqA_{\mu}$$

4. Interpretation

4.1 One Entity, Many Appearances

The field is the visible projection of one entity's full path structure.

4.2 Interference as Path Overlap

Double-slit interference arises from overlapping contributions of the same entity's paths.

4.3 Measurement as Path Selection

Collapse = selecting a subset of paths, not destroying the entity.

4.4 Electrons and Positrons

Different time directions of the same worldline.

4.5 Spin, Charge, and Fields

All standard QFT structure is preserved.

4.6 Curved Spacetime

SEFI extends naturally to gravitational backgrounds.

5. Speculative Extensions (Optional)

5.1 Golden-Ratio Path Weighting

$$w_\varphi[\gamma] = e^{\frac{i}{\hbar}S[\gamma]} \cdot f_\varphi[\gamma]$$

5.2 Unified Substrate and Consciousness

Philosophical, not physical.

5.3 Ontological Unity and Self-Similarity

Optional geometric refinement.

6. Conclusion

Dutton's Unified SEFI provides a mathematically grounded, ontologically unified interpretation of electrons, positrons, and quantum interference. It preserves all predictions of quantum mechanics and quantum field theory while offering a coherent explanation for particle identity, self-interference, and measurement collapse. Optional geometric and metaphysical extensions are clearly separated from the core physics.

7. References

Feynman, R. P. (1948). Space-time approach to quantum mechanics. Schwinger, J. (1951). On gauge invariance and vacuum polarization. Polyakov, A. M. (1987). *Gauge Fields and Strings*. Weinberg, S. (1995). *The Quantum Theory of Fields*.